

Analyze user behavior & optimize the user workflow using Machine Learning Algorithm

ABSTRACT

For any e-commerce website, it's highly important to pay attention to users behavior. Paying attention to the insights makes the business because analyzing these insights helps to know in which way the website can be improved so that it can be very effective in a way that everyone is comfortable using it and in turn improving the business. For this application, the website is linked to Google Analytics and the required user data is extracted and modified to the desired format to perform analysis. Markov Chains, a machine learning algorithm is used for analyzing the user data that is collected.

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INTRODUCTION

Steps followed:

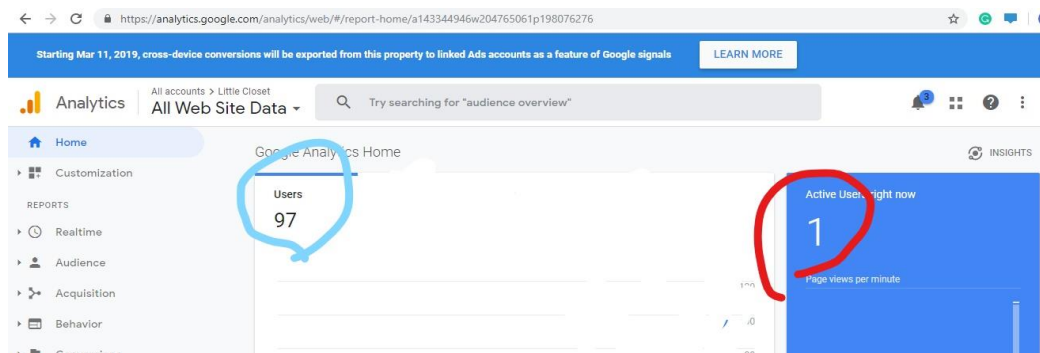
- Website creation
- Data Collection
- Data Preparation
- Applying MarkovClick algorithm to the data
- Analyzing the plots to suggest the changes required

Website Creation

Creating the website is the base for the whole project, this step will impact all the further steps to be followed.

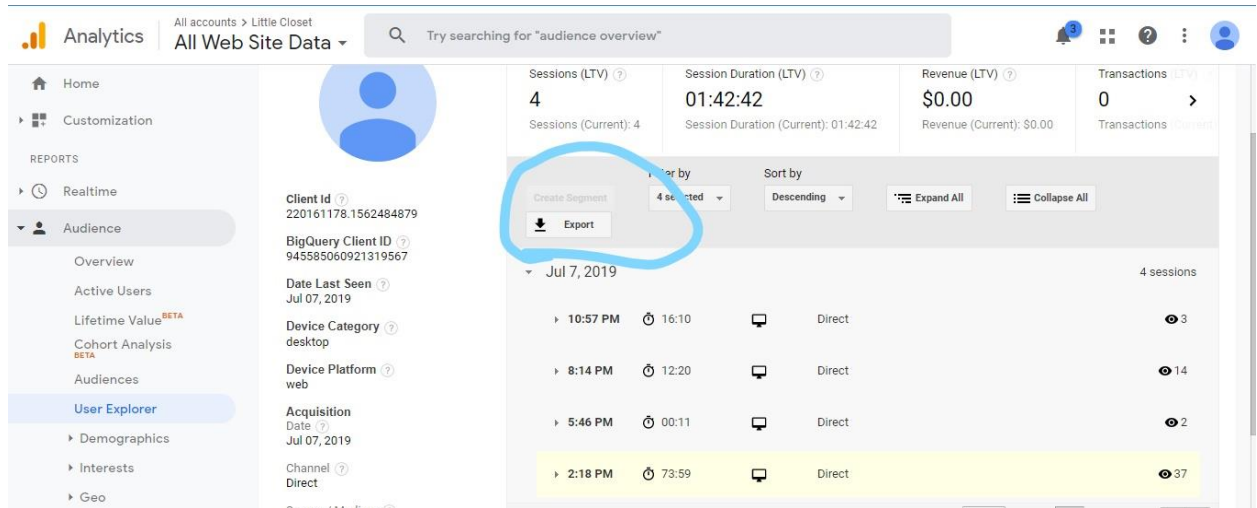
I have used the following steps to create my E-commerce website:

- A free e-commerce website template is chosen that is available in *colorlib*. and several modifications are done to include the required functionalities.
- The website is deployed by drag and dropping the sites folder into *netlify.com*
- *walkme* is used to include the user guide that enables to simplify the online experience and avoid user confusion.
- Include the snippet that is provided by *walkme* in between the head tags in HTML files to make walkme work.
- Connect Google analytics to the website and include the snippet provided by google analytics in all the HTML/PHP files of the site folder.
- Let users surf through the website, so that ample data is collected to perform the required analytics.
- Cross verify if the Google analytics tool is tracking the traffic data, it can be done so by navigating to the home page and checking the user count as shown below.



Data Collection & Preparation

- After making sure that google analytics is tracking our website, navigate to user explorer page where we have to download the information of all the users in JSON format.



The screenshot displays the Google Analytics User Explorer interface. The top navigation bar shows 'Analytics' and 'All Web Site Data'. The left sidebar lists various reports, with 'User Explorer' selected under the 'Audience' section. The main content area shows a user profile for Client ID 220161178.1562484879, with a '4 selected' filter applied. A blue circle highlights the 'Export' button in the top right of the session list. Below this, a table lists sessions for July 7, 2019, with columns for time, duration, device, and channel. The last session is highlighted in yellow.

Time	Duration	Device	Channel	Views
10:57 PM	16:10	Desktop	Direct	3
8:14 PM	12:20	Desktop	Direct	14
5:46 PM	00:11	Desktop	Direct	2
2:18 PM	73:59	Desktop	Direct	37

- All these JSON files are to be converted into a single CSV file by using an online tool or by any other means
- Necessary modifications are done to this CSV file to perform analysis.
- All the Page URLs were assigned with a unique variable, so that everything will be easy to read and an Excel file was generated containing the flow of Page URLs of each session horizontally.

```
In [5]: data=[]
for i in range(79):
    x=df.values[i]
    x=x[~pd.isnull(x)]
    data.append(x)
data
```

```
Out[5]: [array(['A1'], dtype=object),
array(['A2', 'A1'], dtype=object),
array(['A1', 'A1', 'A2', 'A1', 'A2', 'A1', 'A2', 'A1', 'A3', 'A1', 'A3',
'A1', 'A1', 'A3', 'A3', 'A1', 'A1', 'A4', 'A1', 'A1'], dtype=object),
array(['A1', 'A1', 'A3', 'A1', 'A1', 'A3', 'A1', 'A1', 'A1', 'A2', 'A1',
'A2', 'A1', 'A4', 'A1', 'A3', 'A3', 'A1', 'A4', 'A1'], dtype=object),
array(['A1', 'A3', 'A1', 'A3', 'A1', 'A1', 'A3', 'A2', 'A3', 'A3', 'A3',
'A2', 'A3', 'A1', 'A2', 'A2', 'A3', 'A1', 'A3', 'A1', 'A1', 'A3',
'A1'], dtype=object),
array(['A2'], dtype=object),
array(['A2', 'A2', 'A2', 'A1'], dtype=object),
array(['A3', 'A4', 'A3', 'A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1', 'A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1', 'A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1', 'A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1', 'A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A1'], dtype=object),
array(['A2', 'A1'], dtype=object),
```

	A	B	C	D	E	F
1	1 /		A1		/	A1
2	2 /		A1		/category.html	A2
3	3 /category.html		A2		/product.html	A3
4	3 /		A1		/cart.html	A4
5	4 /		A1			
6	4 /		A1			
7	4 /category.html		A2			
8	4 /		A1			
9	4 /category.html		A2			
10	4 /		A1			
11	4 /category.html		A2			
12	4 /		A1			
13	4 /product.html		A3			
14	4 /		A1			
15	4 /product.html		A3			
16	4 /		A1			
17	4 /		A1			
18	4 /product.html		A3			
19	4 /product.html		A3			
20	4 /		A1			

[illegible]

MARKOV CHAINS ALGORITHM

Markovclick allows you to model clickstream data from websites as Markov chains, which can then be used to predict the next likely click on a website for a user, given their history and current state. Markov Chain algorithm is a random process where the next state is dependent on the previous state.

Clickstream is the list of clicks made by each user. This clickstream data is used to model Markov chains with the help of a probability matrix that is formed from the data that has been provided.

Markov chains are mathematical systems that hop from one "state" (a situation or set of values) to another.

A simple, two-state Markov chain is shown below.



Installation:

pip install markovclick

pip install Graphviz

Requirements:

Python 3.X

matplotlib

numpy, pandas

seaborn

```
array(['A2', 'A1'], dtype=object),  
array(['A1'], dtype=object),  
array(['A3', 'A1'], dtype=object),  
array(['A2', 'A1'], dtype=object)]  
  
In [16]: from markovclick.models import MarkovClickstream  
m = MarkovClickstream(data)
```

Data Analytics

Markov analysis was performed on the obtained data and a heat-map was plotted.

Input:

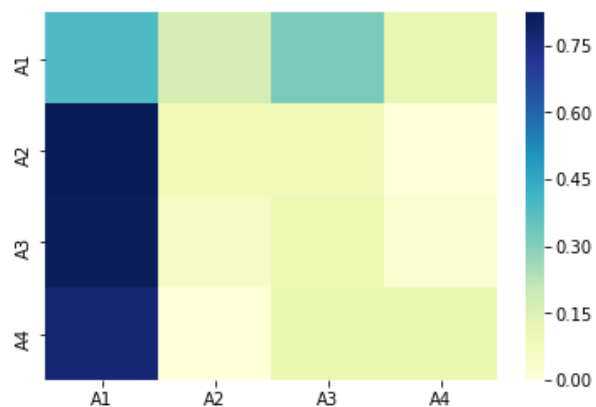
```
from markovclick.models import MarkovClickstream
m = MarkovClickstream(data)
In [19]: sns.heatmap(m.prob_matrix, xticklabels=m.pages, yticklabels=m.pages, cmap="YlGnBu")
```

Output:

Th

```
In [7]: sns.heatmap(m.prob_matrix, xticklabels=m.pages, yticklabels=m.pages, cmap="YlGnBu")
```

```
Out[7]: <matplotlib.axes._subplots.AxesSubplot at 0x1a1b121e48>
```



Insights:

If we observe the above figure, a brief analysis we can make is

- A customer is most likely to come back to home page(A1) after visiting A2 or A3 or A4
- The probability that the user visiting the pages A2 or A3 after visiting A4 is very less
- The users are likely to visit A3(viewing products) from the home page.

Markov chain was plotted for better visualization.

Input:

```
from markovclick.viz import visualise_markov_chain
graph = visualise_markov_chain(m)
graph
```

Output:

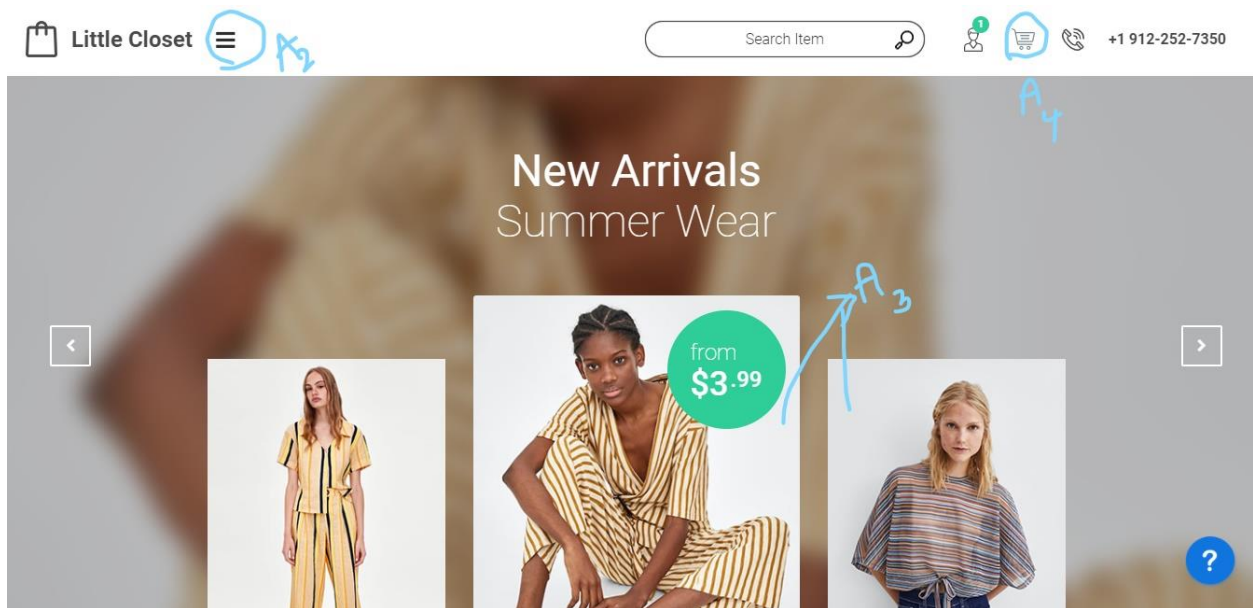
```
In [8]: from markovclick.viz import visualise_markov_chain
graph = visualise_markov_chain(m)
graph
```

Out[8]:



fig: visualizing

ANALYSIS AND SUGGESTED IMPROVEMENTS

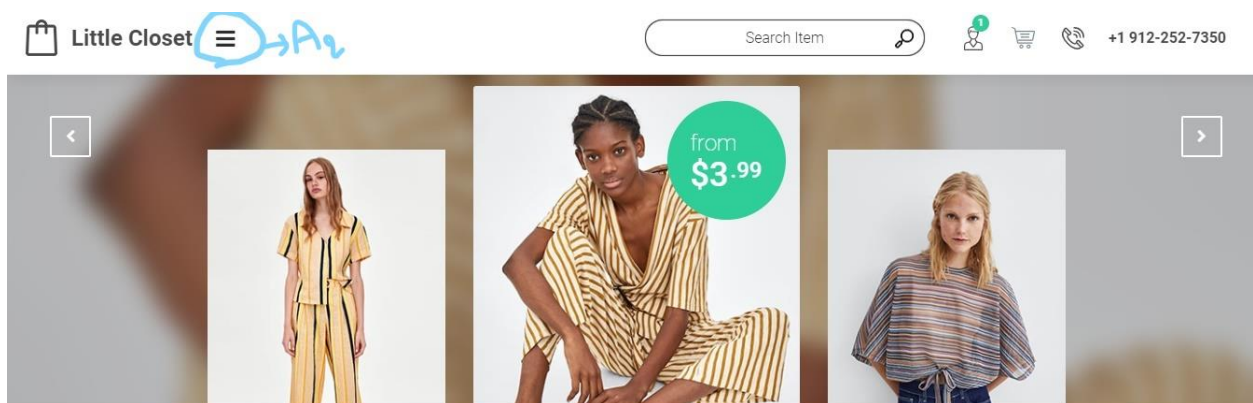


A1 :Home page

A2:categories

A3: viewing details about the products

A4:cart



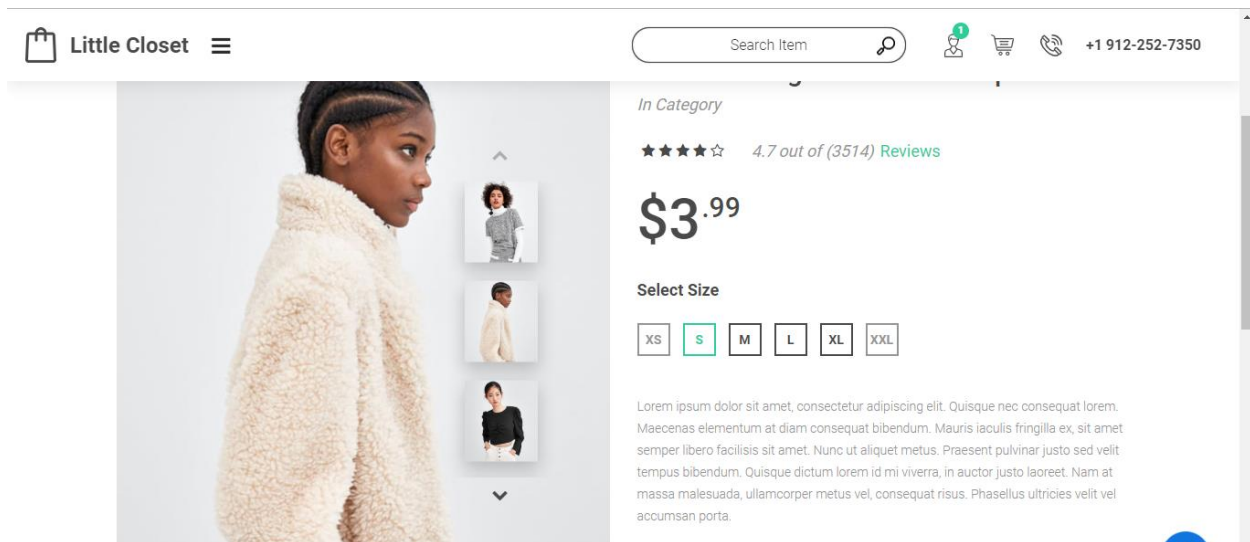
Popular on Little Closet

[Women](#) / [Men](#) / [Kids](#) / [Home Deco](#)

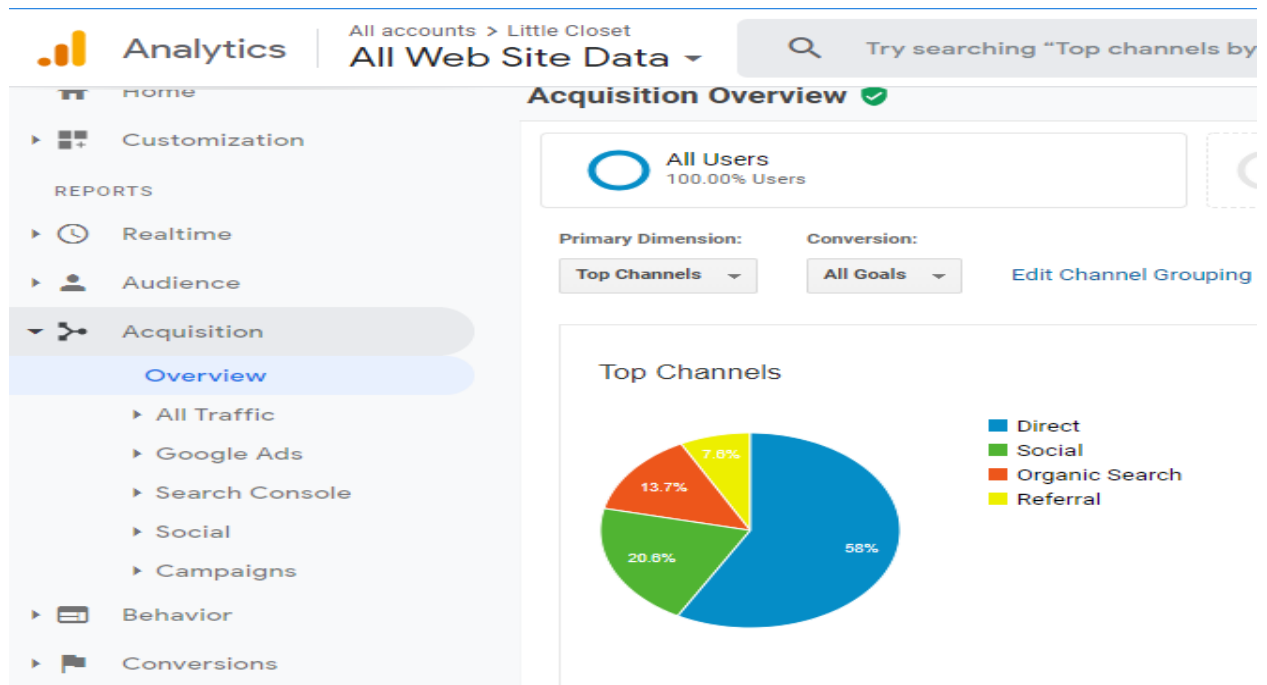
- It is clear from the plot that the user is most likely to get back to the home page after visiting A2 or A3 or A4 . It might be happening because ,most of the products

were already mentioned on the home page so they might not feel the need to visit other pages.

So If only the important and special products are shown on home page ,users may visit various pages to view all products and avail all the services



- The probability that the user navigating from product page to another page (Ax) other than the home page is very less.
This situation can be improved by suggesting the user various other products just below that.



- From the above information, it is important to know that there is less user acquisition through social, organic search, referral categories. So we can suggest that the efforts are needed to improve the user acquisition through these.
- It can be viewed from the Google Analytics that the bounce rate is very high. So we need to improve the home page by adding most popular products, a very good user-friendly design, etc in order to retain the customer.